

FDM for DGS

Step 1. Gauss-Seidel smoothing of u, v

(initial guess of u, v, p can be all zeros)

$$\begin{matrix} u^k \\ v^k \\ p^k \end{matrix} \xrightarrow{G-S} \begin{matrix} u^{k+\frac{1}{2}} \\ v^{k+\frac{1}{2}} \\ p^k \end{matrix}$$

this step is equivalent to ①. relax the momentum equation.

$$\text{i.e. } u^{k+\frac{1}{2}} = u^k + \hat{A}^{-1} (f - \underbrace{\hat{A}u^k}_{\Delta} - \underbrace{\hat{B}^T p^k}_{\nabla})$$

Δ Laplacian operator

∇ gradient operator

G-S

Step 2. Form the residual for pressure

this step is part of ②.

$$r_p = g - \underbrace{\hat{B}u^{k+\frac{1}{2}}}_{\text{div. divergence operator}}$$

div. divergence operator

Step 3.

Solve the Poisson eq. for correction e_p .

with step 2, this step is the end of

② relax the mass eq. i.e.

$$e_p = \hat{A}_p^{-1} r_p$$

this is just regular Poisson $\Delta e_p = r_p$

with Neumann boundary condition.

Step 4, step 5

update u, v, p

this is ③ distribution process.

$$u^{k+1} = u^{k+\frac{1}{2}} + \underline{B}^T e_p$$

$$v^{k+1} = v^{k+\frac{1}{2}} + \underline{B}^T e_p$$

$\underline{B}^T \cdot \nabla$ gradient operator
but in 2 different directions.

$$p^{k+1} = p^k - \underline{B \underline{B}^T} e_p$$

$A_p \cdot \Delta$ Laplacian operator

$$u = 20xy^3$$

$$u_x = 20y^3 \quad u_y = 60xy^2$$

$$u_{xx} = 0$$

$$\cancel{u_{yy}} = 120xy$$

$$\Delta u = u_{xx} + u_{yy} = 120xy$$

$$-\Delta u = -120xy$$

$$v = 5x^4 - 5y^4$$

$$v_x = 20x^3$$

$$v_{xx} = 60x^2$$

$$v_y = -20y^3$$

$$v_{yy} = -60y^2$$

$$\Delta v = v_{xx} + v_{yy} = 60x^2 - 60y^2$$

$$-\Delta v = -60x^2 + 60y^2$$

$$p = 60x^2y - 20y^3$$

$$p_x = 120xy$$

$$p_y = 60x^2 - 60y^2$$

$$\checkmark f_1 = -\Delta u + p_x = -120xy + 120xy = 0$$

$$\checkmark f_2 = -\Delta v + p_y = -60x^2 + 60y^2 + 60x^2 - 60y^2 = 0$$

$$\checkmark g = u_x + v_y = 20y^3 - 20y^3 = 0$$